

Output and Costs

Based on Ch. 11, Economics 10th ed., Parkin, Powell & Mathews

Course: Eco 101 (Introduction to Microeconomics)

Time Frames for Decisions

To study the relationship between a firm's output decision and its costs, we distinguish between two decision time frames:

Short run: In the short run, the quantity of at least one factor of production is fixed and the quantities of the other factors of production can be varied.

Long run: In the long run, the quantities of all factors of production can be varied.

Short-Run Technology Constraint

To increase output in the short run, a firm must increase the quantity of labour employed. We describe the relationship between output and the quantity of labour employed by using **three** related concepts:

1. Total product
2. Marginal product
3. Average product

These product concepts can be illustrated either by product schedules or by product curves.

Product Schedule

- **Total Product (TP):** Total output produced by a given number of workers.
- **Marginal Product (MP):** The additional output from hiring one more worker.

Example: For example, when labor increases from 2 to 3 workers a day (row C to row D), total product increases from 10 to 13 sweaters a day. The marginal product of going from 2 to 3 workers is 3 sweaters

- **Average Product (AP):** Total product divided by number of workers.

Example: The average product of 3 workers is 4.33 sweaters per worker (13 sweaters a day divided by 3 workers).

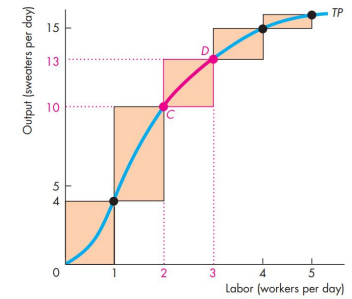
TABLE 11.1 Total Product, Marginal Product, and Average Product

	Labor (workers per day)	Total product (sweaters per day)	Marginal product (sweaters per additional worker)	Average product (sweaters per worker)
A	0	0		
B	1	4	4	4.00
C	2	10	6	5.00
D	3	13	3	4.33
E	4	15	2	3.75
F	5	16	1	3.20

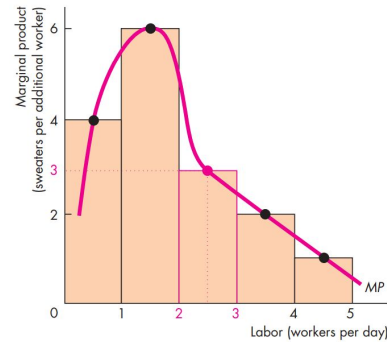
Product Curves

- **Total Product Curve** :A total product curve shows the quantity a firm can produce with a given quantity of capital and different quantities of labor.
- **Marginal Product Curve**: Initially, the marginal product of labor *increases* as the quantity of labor increases, because of increased specialization and the division of labor. Eventually, marginal product *diminishes* because of **the law of diminishing returns**.
- **Average Product Curve**: Initially, average product increases as the quantity of labor increases but eventually average product diminishes.

Average product is largest when average product and marginal product are equal, and the marginal product curve cuts the average product curve at the point of maximum average product. For employment levels at which the marginal product exceeds average product, average product is increasing. For employment levels at which marginal product is less than average product, average product is decreasing.

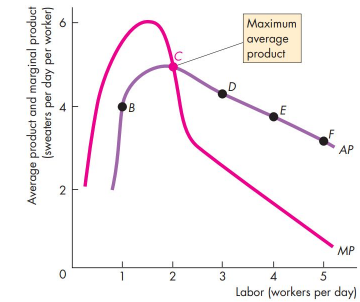


(a) Total product



(b) Marginal product

FIGURE 11.3 Average Product



Law of Diminishing Marginal Returns

Law of Diminishing Marginal Returns: Diminishing marginal returns occur when the marginal product of an additional worker is less than the marginal product of the previous worker.

Diminishing marginal returns arise from the fact that more and more workers are using the same machinery in the same plant. As more workers are added, there is less and less for the additional workers to do that is productive.

For example, from table 11.1 from the previous slide, if the firm employs a third worker, output increases but not by as much as it did when he added the second worker. In this case, after two workers are employed, all the gains from specialization and the division of labour have been exhausted. By employing a third worker, the factory produces more jumpers, but the equipment is being operated closer to its limits. There are even times when the third worker has nothing to do because the plant is running without the need for further attention.

This phenomenon is such a pervasive one that it is called '*the law of diminishing returns*'. The law of diminishing returns states that:

As a firm uses more of a variable input, with a given quantity of fixed inputs, the marginal product of the variable input eventually diminishes.

Short-Run Cost

To produce more output in the short run, a firm must employ more labor, which means that it must increase its costs. We describe the relationship between output and cost by using **three** cost concepts:

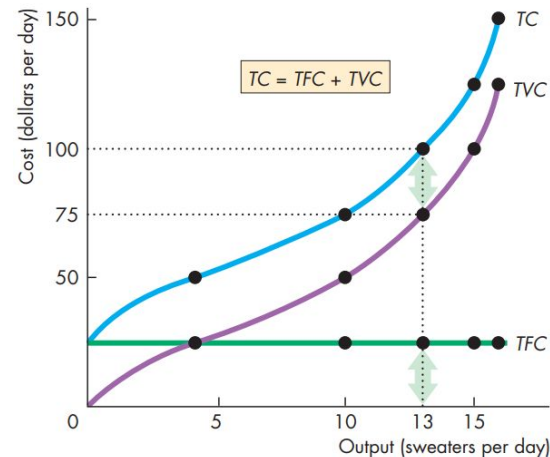
- **Total cost**
- **Marginal cost**
- **Average cost**

Total Cost: A firm's total cost (TC) is the cost of all the factors of production it uses. We separate total cost into **total fixed cost** and **total variable cost**.

- **Total fixed cost (TFC)** is the cost of the firm's fixed factors. The quantities of fixed factors don't change as output changes, so total fixed cost is the same at all outputs.
- **Total variable cost (TVC)** is the cost of the firm's variable factors. Total variable cost changes as output changes.
- **Total cost is the sum of total fixed cost and total variable cost.**

That is, **$TC = TFC + TVC$**

FIGURE 11.4 Total Cost Curves



	Labor (workers per day)	Output (sweaters per day)	Total fixed cost (TFC)	Total variable cost (TVC)	Total cost (TC)
			(dollars per day)		
A	0	0	25	0	25
B	1	4	25	25	50
C	2	10	25	50	75
D	3	13	25	75	100
E	4	15	25	100	125
F	5	16	25	125	150

Marginal Cost: A firm's marginal cost is the increase in total cost that results from a one-unit increase in output.

$$MC = \Delta TC \div \Delta Q$$

Average Costs: Three average costs of production are:

1. **Average fixed cost:** Total fixed cost per unit of output

$$AFC = TFC \div Q$$

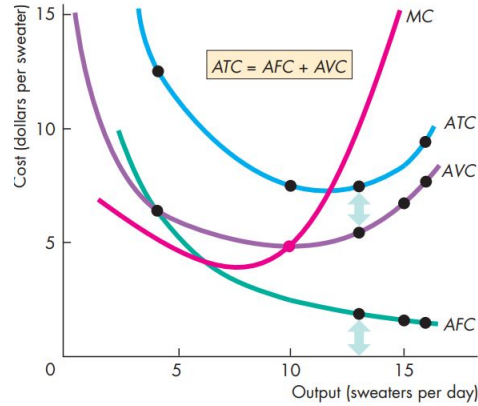
2. **Average variable cost:** Total variable cost per unit of output

$$AVC = TVC \div Q$$

3. **Average total cost:** Total cost per unit of output

$$ATC = AFC + AVC$$

FIGURE 11.5 Marginal Cost and Average Costs



Marginal cost is calculated as the change in total cost divided by the change in output. When output increases from 4 to 10 sweaters, an increase of 6 sweaters, total cost increases by \$25. Marginal cost is $\$25 \div 6$, which is \$4.17.

Each average cost concept is calculated by dividing the related total cost by output. When 10 sweaters are produced, AFC is \$2.50 ($\$25 \div 10$), AVC is \$5 ($\$50 \div 10$), and ATC is \$7.50 ($\$75 \div 10$).

The graph shows that the MC curve is U-shaped and intersects the AVC curve and the ATC curve at their minimum points. The average fixed cost curve (AFC) is downward sloping. The ATC curve and AVC curve are U-shaped. The vertical distance between the ATC curve and the AVC curve is equal to average fixed cost, as illustrated by the two arrows.

	Labor (workers per day)	Output (sweaters per day)	Total fixed cost (TFC)	Total variable cost (TVC)	Total cost (TC)	Marginal cost (MC) (dollars per additional sweater)	Average fixed cost (AFC)	Average variable cost (AVC)	Average total cost (ATC)
			(dollars per day)				(dollars per sweater)		
A	0	0	25	0	25	 6.25	—	—	—
B	1	4	25	25	50	 4.17	6.25	6.25	12.50
C	2	10	25	50	75	 8.33	2.50	5.00	7.50
D	3	13	25	75	100	 12.50	1.92	5.77	7.69
E	4	15	25	100	125	 25.00	1.67	6.67	8.33
F	5	16	25	125	150		1.56	7.81	9.38

Long-Run Cost

- In the long run, a firm can vary both the quantity of labor and the quantity of capital, so in the long run, **all the firm's costs are variable.**
- The behavior of long-run cost depends on the firm's production function, which is the **relationship between the maximum output attainable and the quantities of both labor and capital.**

The table shows the total product data for four quantities of capital (plant sizes).

- The greater the plant size, the larger is the output produced by any given quantity of labor.
- For a given plant size, the marginal product of labor diminishes as more labor is employed.
- For a given quantity of labor, the marginal product of capital diminishes as the quantity of capital used increases.

TABLE 11.3 The Production Function

Labor (workers per day)	Output (sweaters per day)			
	Plant 1	Plant 2	Plant 3	Plant 4
1	4	10	13	15
2	10	15	18	20
3	13	18	22	24
4	15	20	24	26
5	16	21	25	27
Knitting machines (number)	1	2	3	4

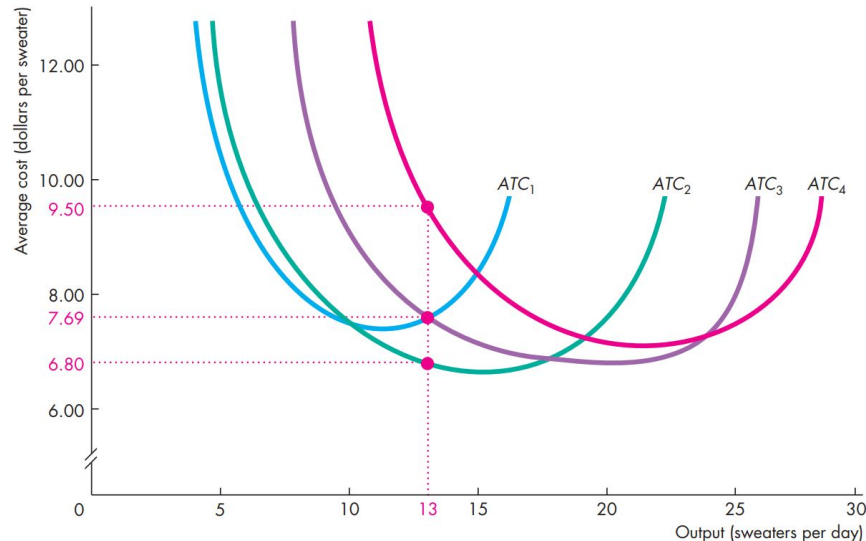
Short-Run Cost and Long-Run Cost

As before, the knitting company can **hire workers for \$25 a day** and **rent knitting machines for \$25 a day**. Using these factor prices and the data in Table 11.3, we can calculate the **average total cost** and graph the ATC curves for factories with 1, 2, 3, and 4 knitting machines.

In Fig. 11.7, two things stand out:

1. Each short-run ATC curve is U-shaped.
1. For each short-run ATC curve, the larger the plant, the greater is the output at which average total cost is at a minimum.

FIGURE 11.7 Short-Run Costs of Four Different Plants



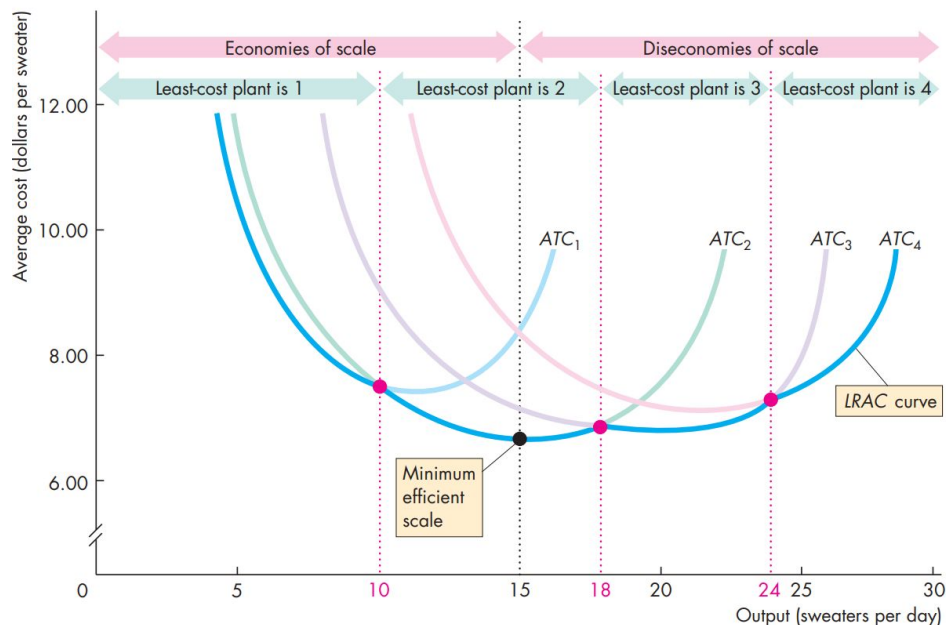
The figure shows short-run average total cost curves for four different quantities of capital at Campus Sweaters. The firm can produce 13 sweaters a day with 1 knitting machine on ATC_1 or with 3 knitting machines on ATC_3 for an average cost of \$7.69 a sweater. The firm can produce 13 sweaters a day by using 2 machines on ATC_2 for \$6.80 a sweater or by using 4 machines on ATC_4 for \$9.50 a sweater.

If the firm produces 13 sweaters a day, the least-cost method of production, the *long-run method*, is with 2 machines on ATC_2 .

Long-Run Average Cost Curve

Definition: A curve that shows the **lowest (unit) cost** at which a firm can produce any given level of output.

FIGURE 11.8 Long-Run Average Cost Curve



The long-run average cost curve traces the lowest attainable ATC when both labor and capital change. The green arrows highlight the output range over which each plant achieves the lowest ATC. Within each range, to change the quantity produced, the firm changes the quantity of labor it employs.

Along the LRAC curve, economies of scale occur if average cost falls as output increases; diseconomies of scale occur if average cost rises as output increases. Minimum efficient scale is the output at which average cost is lowest, 15 sweaters a day.

Economies of Scale, Diseconomies of Scale, and Constant Returns to Scale

Economies of scale are features of a firm's technology that make average total cost fall as output increases. When economies of scale are present, the LRAC curve slopes downward. Greater specialization of both labor and capital is the main source of economies of scale.

Diseconomies of scale are features of a firm's technology that make average total cost rise as output increases. When diseconomies of scale are present, the LRAC curve slopes upward. The challenge of managing a large enterprise is the main source of diseconomies of scale.

Constant returns to scale are features of a firm's technology that keep average total cost constant as output increases. When constant returns to scale are present, the LRAC curve is horizontal.